

Pandas — LEVEL 17: Duplicate Data

Find duplicates, `uplicated()`, `drop_duplicates()`, duplicate rows/columns, keep first/last, and remove all duplicates — line by line with output.

1. Find Duplicates

Duplicates are repeated rows or repeated values in a DataFrame. First, create a DataFrame containing a repeated row.

Code — line by line

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "B", "C"],
    "Age":  [20, 25, 25, 30]
})

print(df)
```

Output

	Name	Age
0	A	20
1	B	25
2	B	25
3	C	30

2. `uplicated()`

`uplicated()` checks whether each row is a duplicate of an earlier row. It returns True for repeated rows and False for the first occurrence.

Code — line by line

```
mask = df.duplicated()

print(mask)
print(df[mask])
```

Output

0	False
1	False
2	True
3	False

dtype: bool

	Name	Age
2	B	25

3. Remove Duplicates

To remove duplicate rows, use `drop_duplicates()`. The first occurrence is kept by default.

Code — line by line

```
result = df.drop_duplicates()

print(result)
```

Output

	Name	Age
--	------	-----

0	A	20
1	B	25
3	C	30

4. drop_duplicates()

`drop_duplicates()` returns a DataFrame after removing duplicate rows. It does not change the original DataFrame unless `inplace=True` is used.

Code — line by line

```
result = df.drop_duplicates()

print("Original:")
print(df)

print("\nAfter removing duplicates:")
print(result)
```

Output

```
Original:
   Name  Age
0    A   20
1    B   25
2    B   25
3    C   30

After removing duplicates:
   Name  Age
0    A   20
1    B   25
3    C   30
```

5. Duplicate Rows

Here we create two completely identical rows. `df.duplicated()` identifies only the later occurrence by default.

Code — line by line

```
df = pd.DataFrame([
    [10, 20],
    [30, 40],
    [10, 20],
    [50, 60]
], columns=["A", "B"])

print(df)

print("\nDuplicate rows:")
print(df[df.duplicated()])
```

Output

```
   A  B
0  10 20
1  30 40
2  10 20
3  50 60

Duplicate rows:
   A  B
2  10 20
```

6. Duplicate Columns

To find duplicate columns, transpose the DataFrame first. After `df.T`, original columns become rows. Then `df.T.duplicated()` can compare them.

Code — line by line

```
df = pd.DataFrame([
    [10, 20, 10],
    [30, 40, 30],
    [50, 60, 50]
], columns=["A", "B", "C"])

print(df)

duplicate_columns = df.T.duplicated()

print("\nDuplicate column mask:")
print(duplicate_columns)

print("\nDuplicate columns:")
print(df.T[duplicate_columns])
```

Output

```
   A  B  C
0 10 20 10
1 30 40 30
2 50 60 50

Duplicate column mask:
A    False
B    False
C     True
dtype: bool

Duplicate columns:
   0  1  2
C 10 30 50
```

7. Keep First

`keep='first'` keeps the first occurrence and marks later duplicates as True. This is the default behavior.

Code — line by line

```
df = pd.DataFrame({
    "Name": ["A", "B", "B", "B", "C"],
    "Age":  [20, 25, 25, 25, 30]
})

print(df.duplicated(keep="first"))
```

Output

```
0    False
1    False
2     True
3     True
4    False
dtype: bool
```

8. Keep Last

`keep='last'` keeps the last occurrence. Earlier occurrences are marked as duplicates.

Code — line by line

```
print(df.duplicated(keep="last"))
```

Output

```
0    False
1     True
2     True
3    False
4    False
dtype: bool
```

9. Remove All Duplicates

To remove every row that belongs to a duplicated group, use `keep=False` with `duplicated()` and keep only rows that are not duplicates.

Code — line by line

```
mask = df.duplicated(keep=False)

print("All duplicate rows:")
print(df[mask])

print("\nDataFrame after removing all duplicate groups:")
print(df[~mask])
```

Output

```
All duplicate rows:
   Name  Age
1    B   25
2    B   25
3    B   25

DataFrame after removing all duplicate groups:
   Name  Age
0    A   20
4    C   30
```

Quick Revision

Task	Code	Meaning
Find duplicate rows	<code>df.duplicated()</code>	True for repeated rows after the first
Remove duplicates	<code>df.drop_duplicates()</code>	Keeps first occurrence by default
Keep first	<code>df.duplicated(keep='first')</code>	Keeps first; marks later duplicates
Keep last	<code>df.duplicated(keep='last')</code>	Keeps last; marks earlier duplicates
Find all duplicate members	<code>df.duplicated(keep=False)</code>	Marks every row in a duplicate group
Remove all duplicate groups	<code>df[~df.duplicated(keep=False)]</code>	Keeps only rows that occur once
Find duplicate columns	<code>df.T.duplicated()</code>	Compares columns after transposing

Important: ``duplicated()`` only identifies duplicates. ``drop_duplicates()`` actually removes duplicate rows from the returned DataFrame.